

# 1 Learning measured-output quantum circuits

Kun and Reyzin [1] studied the complexity of learning a hidden quantum circuit from interactive queries. Their Table 1 considers the four combinations of query type (*function-injection query*, FIQ, versus *value-injection query*, VIQ) and output type (full final state versus a computational-basis measurement). They gave polynomial-query learners for the two FIQ variants and left both VIQ variants open; for measured-output VIQs they additionally showed that the test-path lemma does not apply. We resolve the measured-output VIQ case: its query complexity is exponential in the circuit size. The full-state VIQ problem is not addressed by the argument below.

**Theorem 1.1** (Measured-output VIQ complexity). *Consider the Kun–Reyzin measured-output value-injection model on  $n$  original qubits, all of which are measured at the end, with arbitrary three-qubit gates and value injection into an arbitrary subset of the three output wires of any gate. Let  $Q_{\text{VIQ}}^{\text{meas}}(k, \varepsilon)$  denote the worst-case number of VIQs required to behaviorally  $\ell_1$ -learn every target of size at most  $k$  with success probability at least  $2/3$ . For every fixed  $0 < \varepsilon < \frac{1}{2}$ ,*

$$2^{\Omega(k)} \leq Q_{\text{VIQ}}^{\text{meas}}(k, \varepsilon) \leq O\left(\frac{1728^k}{\varepsilon^2}\right).$$

*In particular, for every fixed  $\varepsilon < \frac{1}{2}$ ,  $Q_{\text{VIQ}}^{\text{meas}}(k, \varepsilon) = 2^{\Theta(k)}$  up to the constant in the exponent.*

The upper bound in Section 1.5 is an exhaustive estimator for arbitrary size- $k$  circuits. The main content is the lower bound in Sections 1.2–1.4: we construct a family  $\{C_s\}_{s \in \{0,1\}^m}$  of size  $O(m)$  whose VIQ responses reveal no more than a constant-gap encoding of one membership query to the hidden point function  $x \mapsto \mathbf{1}[x = s]$ . The mechanism is an *unforgeable quantum token*: during the comparison chain, value injection may destroy the single nonclassical resource created by (G1), but it cannot recreate that resource or move it past a failed comparison.

## 1.1 The model

We use the measured-output model of Kun and Reyzin. A target has  $n$  *original* qubits,  $\ell \leq k$  labeled three-qubit gates  $G_1, \dots, G_\ell$ , and a permutation  $\sigma$  specifying the execution order  $G_{\sigma(1)}, \dots, G_{\sigma(\ell)}$ . The learner is told  $n, \ell$ , and the gate labels, but not the gate matrices, supports, or hidden order. A VIQ addresses gates by these labels. In the lower-bound construction we take  $\sigma$  to be the identity and may reveal all structural information to the learner; this only strengthens the learner. The exhaustive upper bound enumerates the finitely many possible supports and orders.

Each gate is an arbitrary  $8 \times 8$  unitary acting on three original qubits. One- and two-qubit operations are implemented as three-qubit gates by tensoring with identities on dedicated padding qubits. We never use a data qubit as an idle padding wire.

A *value-injection query* (VIQ) specifies:

1. a computational-basis input  $x \in \{0, 1\}^n$ ; and
2. for every labeled gate  $G_j$ , a possibly empty subset  $S_j$  of its three output wires and a string  $v_j \in \{0, 1\}^{|S_j|}$ .

Immediately after  $G_j$  acts, the state on  $S_j$  is discarded and replaced by  $|v_j\rangle$ . On a state  $\rho$  this is the trace-out-and-replace channel

$$\rho \mapsto |v_j\rangle\langle v_j|_{S_j} \otimes \text{Tr}_{S_j}(\rho). \tag{1.1}$$

This channel is operationally equivalent, on the original output qubits, to the fresh-ancilla-and-swap implementation in [1]. The injection subset may be proper and nonempty. In particular, each output wire of a three-qubit gate may independently be left untouched, replaced by  $|0\rangle$ , or replaced by  $|1\rangle$ .

After the last gate and its injection, all  $n$  original qubits are measured in the computational basis. Thus a query  $Q$  returns one sample from a distribution  $\text{Law}(C(Q))$  on  $\{0,1\}^n$ . Fresh ancillas used to implement injection are not part of this measured original register.

The learning goal is behavioral. Given query access to an unknown target  $C$  with at most  $k$  gates, the learner must output a hypothesis  $\hat{C}$  with the same query interface such that

$$\sup_Q \|\text{Law}(\hat{C}(Q)) - \text{Law}(C(Q))\|_1 \leq \varepsilon.$$

The quantity  $Q_{\text{VIQ}}^{\text{meas}}(k, \varepsilon)$  is the worst-case number of VIQs used by a randomized learner that succeeds with probability at least  $2/3$  on every target of size at most  $k$ .

## 1.2 The lower-bound construction $C_s$

Fix  $m \geq 1$  and a hidden string  $s = s_1 \cdots s_m \in \{0,1\}^m$ . The circuit  $C_s$  uses

$$\underbrace{D_1, \dots, D_m}_{\text{data}}, \quad \underbrace{F_0, F_1, \dots, F_m}_{\text{flag chain}}, \quad \underbrace{\tau}_{\text{trap/readout}},$$

together with three dedicated padding qubits  $P_1, P_2, P_3$ . No  $D_i$  is used as padding. The readout qubit  $\tau$  is placed first in the measured register. The gates, in execution order, are as follows.

- (G1) **Create the genuine token.** Apply  $H$  to  $F_0$ , padded by identities on two padding qubits. If the computational input on  $F_0$  is  $r$ , this produces

$$|\chi_r\rangle := H|r\rangle = \begin{cases} |+\rangle, & r = 0, \\ |-\rangle, & r = 1. \end{cases} \quad (1.2)$$

We call the state descended from this particular Hadamard the *genuine token*.

- (G2) **Move the genuine token only along  $s$ .** For  $i = 1, \dots, m$ , apply a controlled-swap  $V_i^{s_i}$  to  $(D_i, F_{i-1}, F_i)$ , where for a computational control  $D_i = a$ ,

$$V_i^b : |a\rangle|u\rangle|w\rangle \mapsto \begin{cases} |a\rangle|w\rangle|u\rangle, & a = b, \\ |a\rangle|u\rangle|w\rangle, & a \neq b. \end{cases} \quad (1.3)$$

Thus  $V_i^{s_i}$  swaps  $F_{i-1}$  and  $F_i$  exactly when  $D_i = s_i$ .

- (G3) **Read out arrival.** Apply  $H$  to  $F_m$ , padded by identities, and then a CNOT from control  $F_m$  to target  $\tau$ , again padded by an identity.
- (G4) **Hide every non-trap wire from the prescribed measurement.** Fix a one-qubit unitary  $R$  satisfying  $R^\dagger Z R = Y$ ; for example,  $R = H S^\dagger$  for  $S = \text{diag}(1, i)$ . Apply  $R$  to every original qubit except  $\tau$ , one wire at a time, padding each operation with identities on dedicated padding qubits.

There are  $n_m = 2m + 5$  original qubits. With the explicit one-wire-at-a-time implementation of (G4), the number of three-qubit gates is

$$k_m = 1 + m + 2 + (n_m - 1) = 3m + 7. \quad (1.4)$$

The constants are inessential; the point is that  $k_m = O(m)$ .

**Lemma 1.2** (Readout asymmetry). *Suppose  $\tau$  enters (G3) in a computational state  $|\tau_0\rangle$ .*

1. *If  $F_m$  holds the genuine token  $|\chi_r\rangle$ , then the readout Hadamard maps it to  $|r\rangle$ , and the CNOT produces  $\tau = \tau_0 \oplus r$  deterministically.*
2. *If  $F_m$  holds a computational-basis state  $|c\rangle$ , then after the readout Hadamard and CNOT the state on  $(F_m, \tau)$  is*

$$|\Phi_{c,\tau_0}\rangle := \frac{|0, \tau_0\rangle + (-1)^c |1, \tau_0 \oplus 1\rangle}{\sqrt{2}}, \quad (1.5)$$

*and the  $\tau$  marginal is maximally mixed.*

*Proof.* The first claim follows from  $H|\chi_r\rangle = |r\rangle$ . For the second,  $H|c\rangle = 2^{-1/2}(|0\rangle + (-1)^c|1\rangle)$ ; applying the CNOT gives (1.5). Tracing out  $F_m$  yields  $I/2$  on  $\tau$ .  $\square$

**Lemma 1.3** (Y-unbiasedness of the hiding layer). *For each of the four states*

$$|0\rangle, \quad |1\rangle, \quad |+\rangle, \quad |-\rangle,$$

*computational-basis measurement after applying  $R$  is uniform on  $\{0, 1\}$ .*

*Proof.* Measuring  $R|\psi\rangle$  in the  $Z$  basis is equivalent to measuring  $|\psi\rangle$  in the  $Y$  eigenbasis, because  $R^\dagger Z R = Y$ . Each of  $|0\rangle, |1\rangle, |+\rangle, |-\rangle$  has squared overlap  $1/2$  with each  $Y$  eigenstate.  $\square$

The hiding layer is a measurement mask, not a state erasure:  $R$  is unitary and therefore preserves the distinguishability of the underlying states. Consequently, this construction does not give a lower bound for an oracle that returns the full final state vector.

### 1.2.1 No-injection behavior: a constant-gap point test

**Proposition 1.4** (Empty VIQ gives a constant-gap point test). *Run  $C_s$  with the empty VIQ on input  $D = x \in \{0, 1\}^m$ ,  $F_0 = 1$ ,  $\tau = 0$ , and all other flag and padding qubits equal to 0. If  $x = s$ , then  $\tau = 1$  deterministically. If  $x \neq s$ , then  $\tau$  is uniform. Every non- $\tau$  output bit is uniform in both cases.*

*Proof.* After (G1), the genuine token is  $|\chi_1\rangle = |-\rangle$  on  $F_0$ . Each  $D_i$  is still the fixed computational input bit  $x_i$  when  $V_i^{s_i}$  acts. Therefore the token moves from  $F_{i-1}$  to  $F_i$  exactly when  $x_i = s_i$ . Once a comparison fails, the token remains on an earlier flag, because all later controlled swaps act only on later flag pairs. Hence the token reaches  $F_m$  exactly when  $x = s$ .

If  $x = s$ , Lemma 1.2(1) gives  $\tau = 1$ , while  $F_m$  is computational before its final  $R$  gate and hence has a uniform measured bit. If  $x \neq s$ , Lemma 1.2(2) makes both the  $\tau$  marginal and the  $F_m$  marginal maximally mixed, so both measured bits are uniform. Every remaining non- $\tau$  wire enters its final  $R$  gate in one of  $|0\rangle, |1\rangle, |+\rangle, |-\rangle$  and is uniform by Lemma 1.3.  $\square$

Thus the empty query gives a constant-gap encoding of membership in the hidden point: a hit produces a deterministic trap bit, while a miss produces a uniform trap bit. It is not literally an exact membership oracle; the exact membership oracle will be used later only as a strengthening in the lower-bound reduction.

### 1.3 Robustness to arbitrary subset injection

The key fact is that the state in (1.2) is the only nonclassical resource created before readout. A value injection can replace it by a computational-basis state, but cannot create another state descended from the token-creation Hadamard.

**Lemma 1.5** (Pre-readout token integrity). *Fix any VIQ  $Q$  on  $C_s$ . From immediately after (G1) until immediately before the readout Hadamard in (G3), after every gate and injection event, the state of the original wires is a product of computational-basis states and at most one genuine token  $|\chi_r\rangle$  on a flag wire. Controlled swaps can only move the genuine token, and value injections can only preserve or destroy it; no value injection can create a genuine token.*

*Proof.* Initially all wires are computational. Gate (G1) creates one genuine token on  $F_0$  and acts as the identity on its padding wires. An injection after (G1) either leaves that token intact or traces it out and replaces it by  $|0\rangle$  or  $|1\rangle$ .

Before  $V_i^{s_i}$  acts, the control  $D_i$  is still its computational input value: no earlier gate touches  $D_i$ , and no data wire is used as padding. Hence  $V_i^{s_i}$  acts as one of two ordinary wire permutations on  $(F_{i-1}, F_i)$ . It may move the genuine token between those two wires, but cannot copy it or create a new one. The state remains a product state. An arbitrary subset injection after  $V_i^{s_i}$  replaces selected factors by computational-basis states; if the selected subset contains the token wire, the genuine token is destroyed. This proves the invariant throughout (G1)–(G2). The readout Hadamard and later gates are deliberately excluded from the statement.  $\square$

For a query  $Q$ , let  $x(Q) \in \{0, 1\}^m$  be its chosen input on  $D_1, \dots, D_m$ . Since  $D_i$  is first touched by  $V_i^{s_i}$ , its value at that comparison is exactly  $x_i(Q)$ ; an injection after  $V_i^{s_i}$  is too late to change that comparison. Define  $\gamma_Q \in \{0, 1\}$  to be 1 exactly when, along the fully matching path, the injection after (G1) does not overwrite  $F_0$  and the injection after each  $V_i^{s_i}$  does not overwrite  $F_i$ . This is a function of the query alone.

**Lemma 1.6** (At most one marked string). *For every VIQ  $Q$  and secret  $s$ , a genuine token is present on  $F_m$  immediately before readout if and only if*

$$\beta_s(Q) := \gamma_Q \mathbf{1}[s = x(Q)] = 1. \quad (1.6)$$

*In particular, a fixed query can mark at most the single candidate  $x(Q)$ .*

*Proof.* If  $s = x(Q)$ , every controlled swap fires and the genuine token follows  $F_0, F_1, \dots, F_m$ . It reaches  $F_m$  exactly when none of the query's injections overwrites the current token wire, which is the event  $\gamma_Q = 1$ . If  $s \neq x(Q)$ , let  $i$  be the first mismatch. The gate  $V_i^{s_i}$  does not swap, so the token, if still present, remains on  $F_{i-1}$ . Every later controlled swap acts on a pair strictly to its right, so the token cannot catch up. If an injection destroys it at any point, Lemma 1.5 shows that it cannot reappear.  $\square$

**Lemma 1.7** (Two-valued joint response). *For every VIQ  $Q$  there exist two distributions  $A_Q$  and  $B_Q$  on  $\{0, 1\}^{n_m}$ , both independent of  $s$ , such that*

$$\text{Law}(C_s(Q)) = \begin{cases} A_Q, & \beta_s(Q) = 1, \\ B_Q, & \beta_s(Q) = 0. \end{cases} \quad (1.7)$$

*Thus the complete joint output distribution, not only its one-bit marginals, depends on  $s$  through the single bit  $\beta_s(Q)$ .*

*Proof.* We condition on the fixed query  $Q$ . Let  $r$  and  $\tau_0$  be the query's computational inputs on  $F_0$  and  $\tau$ .

*State before readout.* By Lemma 1.5, immediately before the readout Hadamard the state is a product. If  $\beta_s(Q) = 1$ , then  $F_m$  holds the genuine token  $|\chi_r\rangle$  and every other wire is computational. If  $\beta_s(Q) = 0$ , then  $F_m$  holds a computational-basis state  $|c\rangle$ ; elsewhere there may be a genuine token on some  $F_j$  with  $j < m$ , or no genuine token at all. The bit  $c$ , the location of a stalled token, and the other computational values may depend on  $s$ .

*The readout pair  $(F_m, \tau)$ .* If the injection immediately after the readout Hadamard overwrites  $F_m$ , then the CNOT receives a query-fixed computational control, so all later behavior of the pair is independent of  $s$ . Suppose it does not overwrite  $F_m$ . When  $\beta_s(Q) = 1$ , the readout Hadamard maps  $|\chi_r\rangle$  to  $|r\rangle$ , and the CNOT produces a computational-basis pair determined by  $Q$ . When  $\beta_s(Q) = 0$ , the pair after the CNOT is  $|\Phi_{c,\tau_0}\rangle$  from (1.5).

First consider no injection into  $F_m$  or  $\tau$  after the CNOT. The final measurement of  $F_m$  after applying  $R$  is a  $Y$ -basis measurement, while  $\tau$  is measured in the  $Z$  basis. For each  $y \in \{+i, -i\}$  and  $t \in \{0, 1\}$ ,

$$\Pr[Y(F_m) = y, Z(\tau) = t \mid \Phi_{c,\tau_0}] = \frac{1}{4}, \quad (1.8)$$

independently of  $c$ : fixing  $t$  selects one of the two computational branches in (1.5), and either  $Y$  outcome has conditional probability  $1/2$ .

An injection after the CNOT into only  $F_m$  fixes  $F_m$  and leaves the maximally mixed  $\tau$  marginal; an injection into only  $\tau$  fixes  $\tau$  and leaves the maximally mixed  $F_m$  marginal; an injection into both fixes both. The same reasoning applies if  $F_m$  is overwritten after its final  $R$  gate: the overwritten output is query-fixed and the remaining  $\tau$  marginal is independent of  $c$ . Therefore the complete measured law of  $(F_m, \tau)$  depends on the secret only through  $\beta_s(Q)$ .

*All remaining wires.* Outside  $(F_m, \tau)$ , the pre-readout state is a product of computational-basis factors and at most one stalled genuine token. Each such wire is either measured after its fixed  $R$  gate, in which case Lemma 1.3 gives a uniform bit, or is overwritten by a later injection, in which case its value is fixed by  $Q$ . Since these wires are product with the readout pair, their full joint law is a product of query-fixed and uniform factors and is independent of their  $s$ -dependent pre- $R$  values or of the stalled token's location.

Consequently, conditioned on  $\beta_s(Q)$ , the entire measured output distribution is a function of  $Q$  alone. Define this distribution to be  $A_Q$  when  $\beta_s(Q) = 1$  and  $B_Q$  when  $\beta_s(Q) = 0$ .  $\square$

**Remark 1.8** (Why the predecessor argument failed, and what fixes it). A natural earlier attempt makes every gate touch  $\tau$ , stores the signal in a  $|\pm\rangle$  phase on  $\tau$ , and argues that any nonempty injection destroys that phase. This is incorrect under arbitrary subset injection: a learner may overwrite only the data outputs of a gate while leaving  $\tau$  untouched, thereby preserving an  $s$ -dependent phase.

The present construction does not rely on every gate touching  $\tau$ . Before readout, the only nonclassical resource is the genuine token created by (G1). A subset injection may destroy that token or modify computational junk, but cannot forge it or advance it past a failed comparison. Moreover, because each  $D_i$  is first touched by its own comparison gate, a query tests exactly its chosen input  $x(Q)$ ; injecting into  $D_i$  after that gate cannot retroactively alter the comparison.

## 1.4 The exponential lower bound

**Lemma 1.9** (Pairwise separation). *For distinct  $s, t \in \{0, 1\}^m$ , the empty query  $Q_s$  from Proposition 1.4, with data input  $D = s$ , separates  $C_s$  and  $C_t$  by  $\ell_1$  distance 1 on the  $\tau$  marginal. Consequently, for every  $\varepsilon < \frac{1}{2}$ , no hypothesis is behaviorally  $\varepsilon$ -close to two distinct members of  $\{C_s : s \in \{0, 1\}^m\}$ .*

*Proof.* Under  $C_s$ , query  $Q_s$  gives the trap marginal  $\delta_1$ . Under  $C_t$ , the same data input is a mismatch and gives  $\frac{1}{2}(\delta_0 + \delta_1)$ . Hence

$$\|\delta_1 - \frac{1}{2}(\delta_0 + \delta_1)\|_1 = 1.$$

If one hypothesis were within  $\varepsilon$  of both targets, the triangle inequality would give a distance at most  $2\varepsilon < 1$  between the two target response distributions on  $Q_s$ , a contradiction.  $\square$

**Theorem 1.10** (Lower bound). *For every fixed  $\varepsilon < \frac{1}{2}$ , any randomized learner that succeeds with probability at least  $2/3$  on every size-at-most- $k$  target requires  $2^{\Omega(k)}$  measured-output VIQs.*

*Proof.* First fix  $m$  and draw  $s$  uniformly from  $\{0, 1\}^m$ . By Lemma 1.7, for every adaptive query  $Q$  the response depends on the secret only through  $\beta_s(Q) = \gamma_Q \mathbf{1}[s = x(Q)]$ , followed by an  $s$ -independent stochastic channel producing  $A_Q$  or  $B_Q$ . Hence an exact membership oracle for the hidden point is at least as informative as the VIQ oracle.

Formally, a simulator answers an adaptive VIQ  $Q$  by querying the membership oracle at  $x(Q)$ . It combines that answer with the query-known bit  $\gamma_Q$  to obtain  $\beta_s(Q)$ , and then samples from  $A_Q$  or  $B_Q$ . Query complexity permits unbounded computation, so these fixed conditional distributions may be computed exactly. Inductively, this simulates the entire adaptive VIQ transcript using one membership query per VIQ.

Now fix all internal coins of a membership learner and all coins used to simulate VIQ responses. The resulting strategy is deterministic. Let  $d \leq q$  be the number of distinct candidates queried along its all-zero transcript. The learner hits the uniformly random secret exactly when the secret is one of these  $d$  candidates, so the hit probability is  $d/2^m$ . Conditioned on no hit, the transcript is that same all-zero transcript for all  $2^m - d$  unqueried secrets. Its fixed output can be  $\varepsilon$ -close to at most one of those secrets by Lemma 1.9. Therefore

$$\Pr[\text{success}] \leq \frac{d}{2^m} + \frac{2^m - d}{2^m} \cdot \frac{1}{2^m - d} = \frac{d + 1}{2^m} \leq \frac{q + 1}{2^m}. \quad (1.9)$$

Averaging over the fixed coins gives the same bound for every randomized membership learner. Hence a learner whose success probability is at least  $2/3$  for every secret must satisfy  $q \geq \frac{2}{3}2^m - 1 = \Omega(2^m)$ .

Finally, the constructed circuits have size  $k_m = 3m + 7$  by (1.4). For a size budget  $k$ , choose  $m = \lfloor (k - 7)/3 \rfloor$  whenever  $k$  is sufficiently large. Then  $k_m \leq k$  and  $2^m = 2^{\Omega(k)}$ .  $\square$

## 1.5 The matching exponential upper bound

**Theorem 1.11** (Upper bound). *For every  $n, k$  and  $0 < \varepsilon < 1$ , the measured-output VIQ complexity satisfies*

$$Q_{\text{VIQ}}^{\text{meas}}(k, \varepsilon) \leq O\left(\frac{1728^k}{\varepsilon^2}\right).$$

*The query bound is independent of  $n$ ; the output length and the unbounded offline computation may depend on  $n$ .*

*Proof.* Let the target have  $\ell \leq k$  gates, and let  $T \subseteq [n]$  be the set of original qubits touched by at least one gate. Then  $t := |T| \leq 3\ell \leq 3k$ .

*Identifying the touched set.* Use input  $1^n$  and, after every gate, inject 000 into all three outputs. For a touched qubit, its last incident gate is followed by an injection that sets it to 0, and no later gate changes it. An untouched qubit remains equal to its input bit 1. Because all original qubits are measured, this one deterministic query reveals  $T$ . Qubits outside  $T$  are always copied unchanged from input to output.

*Enumerating query settings.* Only the input on  $T$  can affect the stochastic part of the response. At each three-qubit gate, each output wire has exactly three choices: untouched, replaced by 0, or replaced by 1. Thus there are  $3^3 = 27$  local injection settings, including the empty one. The number of relevant global settings is therefore at most

$$S \leq 2^t 27^\ell \leq 2^{3k} 27^k = 216^k. \quad (1.10)$$

Proper-subset injections cannot in general be collapsed to full-output injections, because an untouched output may retain quantum information and interact with later gates.

For each setting, the random output is confined to  $T$ , so its alphabet has cardinality

$$M = 2^t \leq 8^k. \quad (1.11)$$

The bits outside  $T$  are deterministic copies of their input bits.

*Estimating all response distributions.* For a distribution  $P$  on an alphabet of size  $M$ , the empirical distribution  $\hat{P}$  from  $N$  independent samples satisfies the standard finite-alphabet bound

$$\Pr[\|\hat{P} - P\|_1 > \eta] \leq (2^M - 2)e^{-N\eta^2/2}.$$

Set  $\eta = \varepsilon/8$  and use a union bound over the  $S$  settings. With overall failure probability at most  $1/3$ , it is enough to take

$$N = O\left(\frac{M + \log(3S)}{\varepsilon^2}\right)$$

samples per setting. The number of oracle calls used for estimation is therefore

$$O\left(\frac{S(M + \log(3S))}{\varepsilon^2}\right) \leq O\left(\frac{1728^k + 216^k k}{\varepsilon^2}\right) \quad (1.12)$$

$$= O\left(\frac{1728^k}{\varepsilon^2}\right), \quad (1.13)$$

where  $SM \leq 27^k 4^{3k} = 1728^k$ . The initial query identifying  $T$  is absorbed into this bound.

*Producing a circuit hypothesis.* It remains to justify that the estimated table can be converted into a size- $\ell$  circuit. Enumerate all possible gate supports inside  $T$ , all labeled execution orders if the order is hidden, and a finite operator-norm  $\zeta$ -net for each three-qubit unitary, where

$$\zeta = \frac{\varepsilon}{16 \max\{1, \ell\}}.$$

This gives a finite class of candidate circuits with the same VIQ interface as the target.

If two corresponding three-qubit gates  $U$  and  $V$  satisfy  $\|U - V\|_{\text{op}} \leq \zeta$ , then for every input state their output density matrices differ in trace norm by at most  $2\zeta$ . A hybrid argument over

the  $\ell$  gates gives trace-norm error at most  $2\ell\zeta \leq \varepsilon/8$ . Between compared gates, both executions apply the same query-dependent trace-out-and-replace channels, later unitaries, partial traces, and measurements; all are trace-distance contractions. Hence a net representative  $C^\star$  of the true target satisfies

$$\sup_Q \|\text{Law}(C^\star(Q)) - \text{Law}(C(Q))\|_1 \leq \frac{\varepsilon}{8}. \quad (1.14)$$

For a candidate  $C'$ , compute exactly, using unbounded offline computation, its response distribution  $P_{C',Q}$  for every one of the  $S$  settings, and minimize

$$d(C') := \max_Q \|P_{C',Q} - \hat{P}_Q\|_1.$$

On the simultaneous estimation event,  $d(C^\star) \leq \varepsilon/4$ , so the selected hypothesis  $\hat{C}$  has  $d(\hat{C}) \leq \varepsilon/4$ . Therefore, for every relevant setting  $Q$ ,

$$\|\text{Law}(C(Q)) - \text{Law}(\hat{C}(Q))\|_1 \leq \|\text{Law}(C(Q)) - \hat{P}_Q\|_1 + \|\hat{P}_Q - \text{Law}(\hat{C}(Q))\|_1 \leq \frac{3\varepsilon}{8} < \varepsilon.$$

The same holds for arbitrary inputs outside  $T$  because both circuits copy those bits unchanged. Thus the learner succeeds with probability at least  $2/3$  using the claimed number of queries.  $\square$

## 1.6 Conclusion

Combining Theorems 1.10 and 1.11 proves Theorem 1.1. Measured-output VIQ learning has exponential query complexity in the circuit size, up to the constant in the exponent and the displayed accuracy factor. The lower bound allows arbitrary adaptive proper-subset injections: each query can only test whether its chosen data input equals the hidden string, because classical value injection can destroy the genuine moving token but cannot recreate it or move it through a failed comparison.

This resolves the measured-output VIQ case left open by Kun and Reyzin. Their Table 1 left both VIQ variants open, not only the measured-output variant. The present construction is specific to measured output: the final  $R$  layer makes the relevant states indistinguishable under the prescribed computational-basis measurement, but because  $R$  is reversible it does not hide the token location from an oracle returning the full state vector.

**Remark 1.12** (Scope). The proof determines the asymptotic dependence on  $k$  only up to constants in the exponent. The explicit upper base 1728 is a conservative consequence of enumerating all 27 arbitrary-subset injection settings per gate and is not claimed to be optimal. The lower-bound family uses  $n_m = \Theta(k_m)$  original qubits. This section does not resolve the full-state VIQ problem.

## References

- [1] Jeremy Kun and Lev Reyzin. Open problem: Learning quantum circuits with queries. In *Proceedings of the 28th Conference on Learning Theory (COLT)*, volume 40 of *PMLR*, pages 1767–1769, 2015.